



**FACULTY OF ENGINEERING
AND TECHNOLOGY**

Course Code: CEF 440

Course Title: Internet Programming (J2EE) and Mobile Programming

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Table of Contents

1. Introduction

2. Tasks Completed This Week

3. Requirements Review and Analysis

- 3.1 Criteria Review Applied (Completeness, Clarity, Feasibility, Dependencies)
- 3.2 Completeness Assessment
- 3.3 Clarity Assessment (Ambiguity Resolution)
- 3.4 Technical Feasibility Assessment
- 3.5 Dependency Analysis

4. Inconsistencies, Ambiguities, and Missing Information

- 4.1 Inconsistencies Found
- 4.2 Ambiguities Found
- 4.3 Missing Information Identified

5. Requirement Prioritisation Summary

- 5.1 Prioritisation Rationale (MoSCoW Method)
- 5.2 Prioritised Requirements Table (Full 50 Requirements)

6. Classification of Requirements

- 6.1 Classification Overview (Functional vs. Non-Functional)
- 6.2 Functional Requirements
- 6.3 Non-Functional Requirements
- 6.4 Classification Notes

7. Highlights from the SRS Document

- 7.1 Core Problem Statement
- 7.2 The Three Content Tiers — Heart of the System
- 7.3 Top 5 User Needs (Survey Evidence)
- 7.4 Technology Stack Summary
- 7.5 Agile Sprint Plan Overview

8. Validation of Requirements with Stakeholders

- 8.1 Validation Methods
- 8.2 Stakeholder Validation Matrix
- 8.3 Sample User Stories for Validation
- 8.4 Validation Questionnaire

1. Introduction

The **Adaptive E-Learning Platform** is being developed to bridge the digital gap for students in Cameroon and similar regions. Most modern educational tools require a fast, stable internet connection, which makes them difficult to use in areas where connectivity is limited. Our goal is to create a platform that works smoothly even when the internet is slow or unreliable.

This report focuses on **Sprint 2**, the initial phase where we gathered and analyzed the project requirements. By consulting with potential users and studying the current market, we have defined the core features needed to make digital learning accessible and inclusive for everyone, regardless of their location or bandwidth.

Summary Item	Detail
Report Period	April 2026 – May 2026 (Week 1 / Sprint 0)
Total Survey Responses	65 respondents (95% students, 83% smartphone users)
Key Problem Confirmed	97% face network problems; 90% have stopped learning due to poor internet
Top Requested Feature	Offline content download (51/59 respondents selected this)
Requirements Gathered	33 Functional + 17 Non-Functional = 50 total requirements
Methodology Adopted	Agile — iterative sprints with flexibility for adjustments
Document Status	SRS v1.0 Draft completed; open for supervisor review
Phase Status	Requirements Gathering — In Progress / Nearing Completion

2. TASKS COMPLETED THIS WEEK

The following tasks were carried out by the team during the requirements gathering and analysis phase. Each task is rated by completion status and linked to the corresponding output.

Task	Output / Deliverable	Status	Owner
Design and distribute requirements survey (21 questions)	Survey form + 59 raw responses	Complete	Team
Clean and organize raw survey data	Cleaned dataset + Data Cleaning Log	Complete	Team

Analyze cleaned data for patterns and needs	Findings tables	Complete	Team
Review existing systems (Kolibri, Khan Academy, etc.)	Existing Systems Review	Complete	Team
Classify Functional and Non-Functional requirements	FR-01 to FR-33 & NFR-01 to NFR-17	Complete	Team
Prioritize requirements (MoSCoW method)	Priority tables (Must/Should/Could)	Complete	Team
Propose system modules and tech stack	5 modules & tech stack table	Complete	Team
Draft Software Requirements Specification (SRS)	SRS Document v1.0 (living draft)	Complete	Team
Design stakeholder validation questionnaire	Finalized questionnaire structure	Complete	Team
Define user stories for key requirements	Sample user stories documented	In Progress	Team
Create clickable prototype / wireframes	Figma / PowerPoint screens	In Progress	Team
Conduct pilot stakeholder validation interviews	Validation Matrix	In Progress	Team

3. REQUIREMENTS REVIEW AND ANALYSIS

This section provides a structured review of the requirements gathered. Each requirement was evaluated against four criteria: Completeness, Clarity, Technical Feasibility, and Dependency relationships.

3.1 Criteria Review Applied

Criterion	What We Checked
Completeness	Does the requirement fully describe what the system must do? Are there gaps or unstated assumptions?
Clarity	Is the requirement written clearly enough that a developer could implement it without guessing?
Technical Feasibility	Can this requirement actually be built given our current tech stack, skill level, and timeline?
Dependencies	Does this requirement depend on another one being completed first? Are those dependencies documented

3.2 Completeness Assessment

We reviewed all 50 requirements (FR-01 to FR-33, NFR-01 to NFR-17) for completeness. Below is a summary of findings:

- **COMPLETE:** Core requirements (adaptive streaming, offline access, user authentication, auto-save) are well-defined, directly supported by survey data, and have clear enough descriptions for implementation.
- **PARTIALLY COMPLETE:** Quiz and assessment requirements (FR-24 to FR-27) are described at a high level but lack specifics — for example, what types of questions will be

supported (multiple choice, open-ended, true/false), how answers will be graded automatically, and what the quiz results screen will show.

- **PARTIALLY COMPLETE:** Communication requirements (FR-28 to FR-30) describe that chat and notifications should exist but do not specify whether chat will be real-time (socket-based) or asynchronous (message queue). This needs clarification before implementation.
- **INCOMPLETE:** Progress Tracking (FR-31, FR-32) does not specify what metrics will be tracked (e.g., time spent per module, percentage watched, quiz scores) or what the dashboard will visually show. These need to be defined in Sprint 1.
- **INCOMPLETE:** Multilingual support (NFR-15, French language) is listed as a Should Have requirement but has no defined scope — for example, whether only the UI will be translated or also the course content. This needs clarification.

3.3 Clarity Assessment

Requirements were assessed for clarity. The following were flagged as ambiguous or potentially confusing:

Requirement ID	Ambiguity Identified	Recommended Action
FR-14	'Good, Fair, or Poor' speed tiers are defined conceptually but specific bandwidth thresholds (e.g., < 500 kbps = Poor) are not set. Different interpretations by team members could lead to inconsistent implementation.	Define numerical thresholds in Sprint 1 (e.g., Poor: < 500 kbps, Fair: 500–2000 kbps, Good: > 2000 kbps).
FR-23	The 'Night Shift' background download is described as occurring at 2:00 AM, but this is an example, not a firm specification. The actual trigger condition is 'strong WiFi signal detected' — time is not the real trigger.	Clarify in SRS: trigger = WiFi signal strength above threshold, not a fixed time. Time-based is optional.
NFR-02	The 3-second load time target is a reasonable estimate but has not been tested. It may not be achievable on the oldest target devices under 3G.	Mark as a target to be validated during Sprint 3 performance testing, not a hard requirement at this stage.
NFR-12	'Last-Write-Wins' conflict resolution is stated as a strategy but it has not been defined whether this applies to quiz answers, progress markers, or all data types. A quiz answer overwritten incorrectly could	Specify which data types use Last-Write-Wins vs. more careful merge strategies. Escalate to mentor if unsure.

Requirement ID	Ambiguity Identified	Recommended Action
	cause data loss.	
FR-29	Push notifications are listed as a desired feature, but the platform (Android only vs. cross-platform) affects which notification service should be used (FCM for Android, APNs for iOS).	Since we are targeting Android first, specify FCM (Firebase Cloud Messaging) as the notification service for now.

3.4 Technical Feasibility Assessment

Each requirement was assessed for technical feasibility given the team's current skill level, available tools, and project timeline. The following summary was produced:

Feature Area	Feasibility	Main Risk	Mitigation
User Authentication (Moodle API)	High	Moodle API configuration complexity	Use Moodle's REST API with token auth; well-documented
HLS Adaptive Bitrate Streaming	Medium	FFmpeg setup and HLS server config	Use pre-built HLS libraries; simplify to 2 quality tiers initially
Hive Local Storage	High	Learning curve for Dart/Flutter beginners	Hive has excellent documentation; start with basic key-value storage
Offline-First Sync (workmanager)	Medium	Background task handling varies by Android version	Test on multiple devices in Sprint 4; use simple polling as fallback
Speed Categorisation Engine	Medium	Accurate speed measurement on mobile	Use small probe file download + timing; validate in Sprint 3
Quiz Module	High	Designing quiz state management	Start with multiple-choice only; expand question types in later sprints
Push Notifications (FCM)	Medium	Firebase setup and Android permissions	Follow official Flutter Firebase notification guide

Feature Area	Feasibility	Main Risk	Mitigation
Night Shift Background Download	Medium	Android Doze mode restricts background tasks	Use workmanager with battery optimisation exemption request
FFmpeg Video Transcoding	Medium	Server-side setup; requires Node.js knowledge	Use pre-written FFmpeg scripts; process videos manually for prototype
Live Video Conferencing	Low — Excluded	Bandwidth-intensive; conflicts with project goal	Excluded from v1; noted as Won't Have in SRS
AI-Based Recommendations	Low — Deferred	Requires ML expertise beyond current team level	Deferred to future sprint or project extension

3.5 Dependency Analysis

The following dependency relationships were identified between requirements. These inform the order in which features should be built across sprints.

Requirement	Depends On	Notes	
FR-06 (Browse courses)	FR-01, FR-02 (Login)	Cannot browse courses without authentication	
FR-09 (Download offline)	FR-06 (Course access)	Must access course content before downloading it	
FR-20, FR-22 (Auto-save & sync)	FR-21 (Offline storage / Hive)	Sync depends on local storage existing first	
FR-15–FR-17 (Auto quality switch)	FR-12–FR-14 (Speed sensing)	Cannot switch quality without knowing current speed	
FR-18 (HLS streaming)	Server: FFmpeg encoding pipeline	HLS requires video pre-processed into .ts segments on server	
FR-25 (Offline quiz submission)	FR-20 (Auto-save) + FR-22 (Sync)	Quiz answers must be saved locally and synced when online	
FR-29 (Push notifications)	Firestore Cloud Messaging (FCM) setup	Notifications depend on FCM integration being complete	
NFR-07 (AES-256 encryption)	FR-21 (Hive DB setup)	Encryption is a property of Hive configuration; DB must exist first	
FR-23 (Night Shift download)	FR-12 (Connectivity sensing) + Hive	Background download requires knowing WiFi is active and storage exists	

4. INCONSISTENCIES, AMBIGUITIES, AND MISSING INFORMATION

During the review of gathered requirements, several inconsistencies, ambiguities, and gaps were identified. Each item is documented below with a severity level and a recommended action.

4.1 Inconsistencies Found

Inconsistencies are cases where two or more requirements or data points appear to conflict with each other.

	Location	Inconsistency	Resolution
1	FR-14 vs. FR-15–FR-17	The three content tiers (Good / Fair / Poor) are defined by words but no bandwidth numbers are given. Two team members may interpret 'Fair' differently.	Define bandwidth thresholds numerically before Sprint 3 begins (e.g., Poor < 500 kbps).
2	NFR-10 vs. FR-22	NFR-10 states '100% offline continuity' but FR-22 states progress syncs 'when connectivity is restored.' If a student's local data is corrupted before sync, 100% continuity cannot be guaranteed.	Qualify NFR-10: '100% continuity for previously downloaded content under normal device conditions.' Add corruption recovery as a future sprint item.
3	FR-23 (Night Shift timing) vs. FR-12 (connectivity sensing)	FR-23 describes Night Shift as happening at '2:00 AM' but FR-12 describes it as triggered by WiFi signal. These are two different trigger mechanisms.	Consolidate: Night Shift triggers on strong WiFi detection, regardless of time. Time-based scheduling is an optional enhancement.
4	Tech stack vs. simplicity goal	The proposed stack (HLS + Hive + FFmpeg + Moodle + workmanager + FCM + Cloudflare) is comprehensive but may be too complex for beginner developers to implement fully within a single school project.	Apply phased simplification: Sprint 1–2 uses basic REST API + simple video player; adaptive features added in Sprint 3–4 after team gains experience.

4.2 Ambiguities Found

Ambiguities are cases where a requirement or finding could be interpreted in more than one way.

Ambiguity	Why It Is a Problem	Suggested Clarification
'Chat with teacher' (FR-28) — real-time or async?	Real-time chat requires socket connections (complex). Async messaging is simpler but less immediate. The survey just said 'chat with teacher' without specifying.	Define as asynchronous messaging for v1 (messages sent, replied to later). Real-time chat deferred to a later sprint.
'Notifications' (FR-29) — in-app,	Different notification types	Implement in-app notifications

push, or SMS?	require different implementations. The survey listed notifications as a feature without specifying the type.	first (simpler). Add FCM push notifications in Sprint 5 if time allows.
'Quizzes and tests' (FR-24) — what question types?	Multiple choice, short answer, true/false, and image-based questions all have different storage and display requirements.	Limit to multiple-choice only in v1. Other question types documented for future sprints.
'Low-end device' target (NFR-04) — what minimum spec?	Without a minimum hardware specification, we cannot test properly or optimise for the right target. 'Low-end Android' is too vague.	Define minimum spec: Android 6.0+, 2GB RAM, ARMv7 processor. Test on this baseline in Sprint 4.
'Multilingual support' (NFR-15) — UI only or content too?	Translating the UI is feasible. Translating course content added by teachers is impractical to automate.	Scope: translate the UI strings only (English/French toggle). Course content remains in the language the teacher uploaded it in.

4.3 Missing Information Identified

The following information was not available at the time of requirements gathering and needs to be obtained or decided before or during Sprint 1.

Missing Item	Why It Is Needed	How to Obtain
Bandwidth threshold values for speed tiers (Good / Fair / Poor)	Without thresholds, the adaptive logic cannot be implemented precisely.	Research literature and test with local network conditions in Sprint 3.
Moodle server access credentials and API endpoint	Moodle is our proposed backend but we have not confirmed access to a Moodle instance.	Supervisor or IT department to confirm. If unavailable, build simple custom REST API as fallback.
Minimum Android device specification for testing	We need a baseline device to test performance, data usage, and offline behaviour.	Team to define baseline spec (see ambiguity above) and identify a physical device for testing.
Server hosting details (for FFmpeg + Node.js + Moodle)	We need a server to deploy the backend. Free-tier cloud options may have limitations.	Research options: Render, Railway, or Heroku free tier. Confirm with supervisor.
Exact number of students in pilot test group	The validation plan mentions 'a small group of users' but no number is defined.	Aim for 10–20 students from the target user group for pilot testing in Sprint 5–6.
User interface language	Survey showed French	Team decision: start English-only,

Missing Item	Why It Is Needed	How to Obtain
decision (English, French, or bilingual)	responses. App language affects UI design from the beginning.	add French toggle in Sprint 5 if schedule allows.
Whether Moodle quiz module will be used or a custom one built	Determines how quiz data is structured, stored, and synced.	Evaluate Moodle quiz API documentation. If too complex, build basic custom quiz module.

5. REQUIREMENT PRIORITISATION SUMMARY

Requirements were prioritised using the MoSCoW method, evaluated on two dimensions: importance to users (based on survey data) and feasibility for the team (based on technical assessment).

5.1 Prioritisation Rationale

The following criteria were used to assign priorities:

Priority Level	Criteria	Examples
MUST HAVE	Directly solves the core problem (connectivity-resilient learning). Survey evidence is strong (>80% of respondents). Technically feasible for the team.	Adaptive streaming, offline download, auto-save, user login
SHOULD HAVE	Significantly improves user experience. Survey evidence is moderate. Feasible but may require more time.	Manual data saver, quiz module, notifications, progress dashboard
COULD HAVE	Adds value but is not essential. Technically possible but may exceed project timeline.	AI recommendations, gamification, advanced analytics
WON'T HAVE (v1)	Conflicts with project constraints or requires resources beyond scope.	Live video conferencing, third-party payments

5.2 Prioritised Requirements Table (All 50 Requirements)

ID	Requirement	Priority	Sprint Target
FR-01	User registration (student and teacher)	Must Have	Sprint 1
FR-02	Secure user login	Must Have	Sprint 1
FR-06	Browse and access assigned courses	Must Have	Sprint 1
FR-07	Teacher course content upload	Must Have	Sprint 1

ID	Requirement	Priority	Sprint Target
FR-08	Multi-format content display (video / audio / text)	Must Have	Sprint 2
FR-09	Download course materials for offline use	Must Have	Sprint 4
FR-10	Modular course content organisation	Must Have	Sprint 2
FR-11	Assignment submission	Must Have	Sprint 2
FR-12	Real-time connection type detection	Must Have	Sprint 3
FR-13	Active internet verification via ping	Must Have	Sprint 3
FR-14	Download speed measurement and tier classification	Must Have	Sprint 3
FR-15	Auto switch to HD Video when connection is Good	Must Have	Sprint 3
FR-16	Auto switch to SD Video when connection is Fair	Must Have	Sprint 3
FR-17	Auto switch to Lite Mode (audio + text) when Poor	Must Have	Sprint 3
FR-18	HLS Adaptive Bitrate Streaming (10-second segments)	Must Have	Sprint 3
FR-20	Automatic local progress saving (Hive)	Must Have	Sprint 4
FR-21	Full offline access to downloaded content	Must Have	Sprint 4
FR-22	Auto background sync when connectivity restored	Must Have	Sprint 4
NFR-01	HLS to prevent buffering during playback	Must Have	Sprint 3
NFR-07	AES-256 encryption for locally stored data	Must Have	Sprint 4
NFR-10	100% offline continuity (downloaded content)	Must Have	Sprint 4
NFR-13	Simple, intuitive UI for basic digital literacy users	Must Have	Sprint 1
NFR-14	Mobile-responsive layout for smartphone screens	Must Have	Sprint 1
FR-03	Password recovery (email reset)	Should Have	Sprint 2
FR-19	Manual Data Saver toggle (audio-only)	Should Have	Sprint 3
FR-23	Night Shift background download strategy	Should Have	Sprint 4
FR-24	Quizzes associated with course modules	Should Have	Sprint 5

ID	Requirement	Priority	Sprint Target
FR-25	Offline quiz answer saving and sync	Should Have	Sprint 5
FR-26	Quiz results display to students	Should Have	Sprint 5
FR-27	Teacher quiz creation and management	Should Have	Sprint 5
FR-28	Student-to-teacher messaging	Should Have	Sprint 5
FR-29	Push notifications	Should Have	Sprint 5
FR-30	Offline notification queue and delivery	Should Have	Sprint 5
FR-31	Student progress dashboard	Should Have	Sprint 5
FR-32	Teacher view of student progress	Should Have	Sprint 5
NFR-04	Native ARM compilation (low-end device optimisation)	Should Have	Sprint 1
NFR-05	Data consumption minimisation	Should Have	Sprint 3
NFR-06	Ambient UI adaptation (hide images on 2G)	Should Have	Sprint 3
NFR-11	Graceful network loss handling (no crash/freeze)	Should Have	Sprint 3
NFR-15	French language UI support	Should Have	Sprint 7
FR-33	QoE event logging for research evaluation	Could Have	Sprint 6
NFR-12	Last-Write-Wins sync conflict resolution	Could Have	Sprint 4
NFR-16	Modular architecture for future feature addition	Could Have	Sprint 1
NFR-17	Scalable backend (concurrent user support)	Could Have	Sprint 7
–	Live video conferencing	Won't Have (v1)	Excluded
–	AI/ML personalisation engine	Won't Have (v1)	Excluded
–	Gamification (badges, streaks)	Won't Have (v1)	Excluded
–	Third-party payment integration	Won't Have (v1)	Excluded
–	Real-time collaborative editing	Won't Have (v1)	Excluded
–	Social media integration	Won't Have (v1)	Excluded

ID	Requirement	Priority	Sprint Target
–	Advanced analytics dashboard	Won't Have (v1)	Excluded

6. Classification of Requirements

All requirements have been classified as either Functional (what the system does) or Non-Functional (how the system performs). This section provides a summary view and a discussion of each category.

6.1 Classification Overview

Category	Description	Count in This Project
Functional Requirements (FR)	Describe specific behaviours, actions, and features the system must provide. These are the visible capabilities users interact with.	33 (FR-01 to FR-33)
Non-Functional Requirements (NFR)	Describe quality attributes and constraints: how fast, how secure, how reliable, how easy to use the system must be.	17 (NFR-01 to NFR-17)
TOTAL		50 requirements

6.2 Functional Requirements — By Feature Area

Feature Area	Requirement IDs	Number of Requirements
User Authentication & Account Management	FR-01 to FR-05	5
Course Content Access & Management	FR-06 to FR-11	6
Adaptive Content Delivery	FR-12 to FR-19	8
Offline Access & Progress Synchronisation	FR-20 to FR-23	4

Quizzes & Assessments	FR-24 to FR-27	4
Communication & Notifications	FR-28 to FR-30	3
Progress Tracking	FR-31 to FR-33	3
TOTAL		33

6.3 Non-Functional Requirements — By Quality Attribute

Quality Attribute	Requirement IDs	Number of Requirements
Performance	NFR-01 to NFR-03	3
Resource Efficiency	NFR-04 to NFR-06	3
Security	NFR-07 to NFR-09	3
Reliability & Availability	NFR-10 to NFR-12	3
Usability & Accessibility	NFR-13 to NFR-15	3
Scalability	NFR-16 to NFR-17	2
TOTAL		17

6.4 Classification Notes

- The largest functional area is Adaptive Content Delivery (8 requirements), which reflects the core technical challenge and innovation of this project.
- Security requirements (NFR-07 to NFR-09) are all marked Must Have or Should Have because device theft and shared device usage are real risks in the target environment.
- Communication requirements (FR-28 to FR-30) are Should Have, not Must Have, because the system can be used for learning without messaging features. They become important once the platform grows.

- Performance and reliability requirements directly support the project aim: if the system buffers or crashes on a poor connection, it fails its core purpose even if all other features are present.

7. HIGHLIGHTS FROM THE SRS DOCUMENT

This section summarises the most important points from the Software Requirements Specification (SRS v1.0) for quick reference by team members and the supervisor.

7.1 Core Problem Statement

97% of survey respondents face network problems during online learning, and 90% have stopped learning entirely due to poor internet. Existing platforms are built for stable, high-speed connections and fail in low-resource settings like Cameroon. The system must treat poor connectivity as the default, not an exception.

7.2 The Three Content Tiers — Heart of the System

The most critical technical feature of the system is the three-tier adaptive content delivery logic:

Tier	Trigger Condition	Content Mode	Technology
Tier 1 — Good	Download speed > ~2 Mbps	HD Video (720p)	HLS + chewie player
Tier 2 — Fair	~500 kbps – 2 Mbps	SD Video (360p)	HLS + chewie player
Tier 3 — Poor	< ~500 kbps or no signal	Audio + Text (Lite Mode)	audioplayers + text view

7.3 Top 5 User Needs (Survey Evidence)

	User Need	Survey Evidence	SRS Requirement
1	Offline content access	51/59 respondents selected 'Download for offline use'	FR-09, FR-21
2	Automatic content adjustment	97% said Yes or Maybe to adaptive content	FR-12 to FR-18
3	Automatic progress saving	92% want auto-save	FR-20, FR-22
4	Low data usage	27% cited it as the top priority; 44% find data expensive	NFR-05, FR-19
5	Smooth playback (no buffering)	27% cited smooth playback as top priority	NFR-01, FR-18

7.4 Technology Stack Summary

Component	Technology	Purpose
Frontend	Flutter (Dart)	Cross-platform mobile app; native ARM performance
Connectivity	connectivity_plus + pinger	Detect connection type and verify real internet
Video Streaming	HLS + chewie	Adaptive bitrate video; prevents buffering
Local Storage	Hive (NoSQL)	Offline data storage; encrypted with AES-256
Background Tasks	workmanager	Night Shift downloads + offline sync triggers
Backend / LMS	Moodle API + Node.js	Course management and authentication
Media Processing	FFmpeg	Server-side video transcoding to multiple resolutions
Cloud Sync	Firebase / Couchbase Lite	Bridge between Hive and central server
CDN	Cloudflare	Reduces latency for users in Cameroon

7.5 Agile Sprint Plan Overview

Sprint	Focus	Key Deliverables
0 (Now)	Planning & Requirements	SRS v1.0, requirements classification, tech stack decision
1	Auth & Basic UI	Login, registration, course list, mobile layout
2	Content Delivery	Video/audio/text player, manual mode toggle, assignments
3	Adaptive Logic	Speed sensing, auto tier switching, HLS integration
4	Offline & Sync	Hive storage, download, auto-save, background sync
5	Quizzes & Communication	Quiz module, notifications, chat, progress tracking
6	Polish & Pilot Testing	UI refinement, bug fixes, stakeholder pilot test
7 (TBD)	Optional Enhancements	French UI, advanced analytics, gamification (if time allows)

8. VALIDATION OF REQUIREMENTS WITH STAKEHOLDERS

Requirement validation is the process of confirming that the documented requirements actually reflect the needs of the people who will use or be affected by the system. Three validation methods are being used for this project.

8.1 Validation Methods

Method	Description	When / Status
Prototyping & Demo	Clickable mockups created in Figma or PowerPoint showing key screens and adaptive flow scenarios. Shown to supervisor and sample users.	Sprint 0 (In Progress)
Stakeholder Questionnaire	Structured 8-section Google Form sent to students, teachers, and administrators to confirm requirements match real needs.	Sprint 0–1 (Planned)
User Stories + Pilot Testing	Short user stories written for key scenarios. Pilot test with 10–20 real users in Cameroon during Sprint 5–6.	Sprint 5–6 (Planned)

8.2 Stakeholder Validation Matrix

The table below shows the current validation status for each stakeholder group.

Stakeholder	Validation Method	Status	Success Criterion
Students (Primary)	Survey + pilot testing + usability scenarios	Survey: Complete. Pilot: Planned (Sprint 6)	Can continue learning without frustration when internet drops
Teachers	Prototype walkthrough + functional demo	Planned (Sprint 1–2)	Can upload content easily; students can access without errors
Developers (Team)	Technical testing of sensing and switching logic	In Progress (Sprint 3)	System detects connection states and selects correct tier
System Administrators	DB inspection + performance monitoring logs	Planned (Sprint 4–6)	Logs stored correctly; system stable under varying load
Educational Institutions	Pilot testing with student/teacher groups	Planned (Sprint 6)	Feedback shows improved learning continuity

Stakeholder	Validation Method	Status	Success Criterion
Course Supervisor	SRS document review + prototype demonstration	In Progress (Sprint 0)	SRS is complete, clear, and feasible for a school project

8.3 Sample User Stories for Validation

The following user stories have been drafted for key requirements. These will be used in the validation questionnaire and pilot test.

	Stakeholder	User Story
1	Student	As a student with poor mobile data, I want the app to automatically switch to audio-only mode so I can continue listening to my lecture without the video buffering.
2	Student	As a student in a rural area, I want to download my course notes when I have WiFi at school so I can study at home where there is no internet.
3	Student	As a student, I want the app to save my quiz answers automatically so that if my internet cuts out mid-quiz, I do not lose my progress.
4	Teacher	As a teacher, I want to upload a lecture video and have the system automatically prepare a low-bandwidth version so that students with weak connections can still watch it.
5	Teacher	As a teacher, I want to see how far each student has progressed in my course so I can identify who is falling behind.
6	Developer	As a developer, I want the connectivity module to accurately classify the network speed so that the app selects the correct content tier without switching too frequently.
7	Admin	As a system administrator, I want the app to log when and why content quality was reduced so I can analyse the Quality of Experience data.
8	Institution	As an educational institution, I want the platform to work reliably even in areas with 2G coverage so that no student is excluded from learning due to their location.

8.4 Validation Questionnaire

The following is a summary of the 8-section validation questionnaire that will be distributed to stakeholders in Sprint 1. The full question list is included in Appendix A of this report.

Section	Topic	Focus Areas
1	User Management	Need for sign-up/login, importance of secure authentication
2	Course Access	Preferred content types, quiz usage frequency
3	Adaptive Streaming	Value of automatic quality adjustment, preferred fallback format
4	Network Monitoring	Importance of buffering prevention, latency awareness
5	Content Flexibility	Manual format selection preference, offline format priority
6	Personalisation	Value of content recommendations based on past activity
7	Offline Support	Importance of offline access, which content types to prioritise offline
8	Non-Functional	Data minimisation, device compatibility, reliability, security

9. ISSUES, RISKS, AND ACTION ITEMS

9.1 Open Issues

#	Issue	Severity	Owner	Target Sprint
1	Bandwidth thresholds for speed tiers not yet defined — adaptive logic cannot be implemented without these.	High	Team	Sprint 1
2	Moodle server access not confirmed — fallback REST API plan needs to be prepared.	High	Team + Supervisor	Sprint 1
3	Minimum Android device spec not defined — cannot optimise or test without a baseline.	Medium	Team	Sprint 1
4	Quiz requirement scope too broad — question types, grading logic, and results format not specified.	Medium	Team	Sprint 2
5	Night Shift trigger ambiguity — time-based vs. signal-based trigger not resolved.	Medium	Team	Sprint 2

#	Issue	Severity	Owner	Target Sprint
6	Multilingual scope not defined — UI only or content too?	Low	Team	Sprint 3
7	Pilot test group size and recruitment not yet planned.	Low	Team	Sprint 4
8	Server hosting for backend not selected — free tier may have limitations.	Medium	Team	Sprint 1

9.2 Risks

Risk	Likelihood	Impact	Mitigation Strategy
HLS + FFmpeg implementation is too complex for beginner team members.	Medium	High	Simplify to 2 quality tiers in Sprint 3; use pre-built HLS libraries; consult supervisor.
Moodle API unavailable or too complex to integrate.	Medium	High	Build simple custom Node.js REST API as fallback; treat Moodle as a future enhancement.
Android background task restrictions prevent Night Shift downloads.	Medium	Medium	Use workmanager's battery-optimisation exemption request; test on physical device in Sprint 4.
Team velocity lower than expected — sprint targets not met.	Medium	Medium	Agile plan allows for feature de-scoping. Move lower-priority items to later sprints.
Survey data (59 respondents) may not fully represent all user needs.	Low	Low	Supplement with user story interviews in Sprint 5; adjust requirements as needed.
Server costs exceed free-tier limits during testing.	Low	Medium	Use Render, Railway, or Heroku free tier; limit test uploads to small files during development.

9.3 Action Items for Next Sprint

#	Action Item	Owner	Priority	Sprint
1	Define numerical bandwidth thresholds for Good / Fair / Poor speed tiers.	Team	High	Sprint 1
2	Confirm Moodle server access or	Team +	High	Sprint 1

#	Action Item	Owner	Priority	Sprint
	commit to custom REST API fallback.	Supervisor		
3	Define minimum Android device specification (e.g., Android 6.0+, 2GB RAM).	Team	High	Sprint 1
4	Set up Flutter project structure with Hive, connectivity_plus, and basic navigation.	Team	High	Sprint 1
5	Build login and registration screens connected to backend (Moodle or custom API).	Team	High	Sprint 1
6	Create course list screen with basic content display (video / audio / text tabs).	Team	High	Sprint 1
7	Distribute stakeholder validation questionnaire and collect responses.	Team	Medium	Sprint 1
8	Refine user stories (expand beyond the 8 samples) and get supervisor sign-off.	Team	Medium	Sprint 1
9	Select server hosting option and set up basic backend environment.	Team	Medium	Sprint 1
10	Complete prototype / wireframes for the adaptive flow demo.	Team	Medium	Sprint 1
11	Update SRS document based on supervisor and peer review feedback.	Team	Medium	Sprint 1

10. NEXT WEEK PLAN (SPRINT 1 PREVIEW)

Sprint 1 will shift from planning and requirements to building the first working components of the application. The focus will be on authentication, basic course navigation, and completing the outstanding validation activities from Sprint 0.

#	Planned Task	Deliverable	Priority	Days Est.
1	Finalise bandwidth threshold values for speed tiers.	Updated SRS Section 3.3	High	1
2	Set up Flutter project with routing, state management, and Hive integration.	Boilerplate app structure	High	2
3	Build login and registration screens (connected to Moodle or custom API).	Working auth flow	High	3

#	Planned Task	Deliverable	Priority	Days Est.
4	Build course list / home screen.	Course browsing UI	High	2
5	Set up basic content viewer (video, audio, text tabs).	Multi-format content view	High	2
6	Distribute validation questionnaire; collect and summarise responses.	Validation summary report	Medium	2
7	Finalise and expand user stories (all 8 stakeholder categories).	Full user story document	Medium	1
8	Complete prototype / Figma wireframes for adaptive flow demo.	Clickable prototype	Medium	2
9	Research and select server hosting option.	Backend environment live	Medium	1
10	Update SRS based on supervisor feedback received after Sprint 0.	SRS v1.1	Medium	1

KEY SURVEY DATA POINTS

The following table summarises the most cited survey findings that directly drove requirements decisions in this report.

Finding	Value	Requirement Influenced
Students among respondents	95%	User type focus — system primarily for students
Smartphone users (exclusively or combined)	83%	NFR-14: Mobile-first design mandatory
Users with poor or average internet (rating $\leq 3/5$)	95%	FR-12 to FR-18: Adaptive delivery is the core feature
Users who face network problems always or sometimes	97%	NFR-10, NFR-11: Reliability and continuity essential
Users relying on mobile data	83%	NFR-05: Data minimisation required
Users who find data expensive or just manageable	91%	FR-19: Manual Data Saver; FR-23: Night Shift
Users wanting automatic content adjustment (Yes + Maybe)	97%	FR-14 to FR-18: Auto adaptive switching is a must
Users selecting offline download as desired feature	51/59 (86%)	FR-09, FR-21: Offline access is the top feature

Finding	Value	Requirement Influenced
Users rating offline access 4 or 5 out of 5	80%	FR-21: Must Have priority confirmed
Users wanting automatic progress saving	92%	FR-20, FR-22: Auto-save and sync are required
Users who have stopped learning due to poor internet	90%	Confirms the core problem the system must solve
Most preferred fallback (text notes)	56%	FR-17: Lite Mode must include text; FR-08: text is mandatory
Top factor when learning online (tie)	27% low data + 27% smooth playback	NFR-05, NFR-01: Both equally critical

Conclusion

The survey results show a massive gap between current technology and the reality for students. Since **97% of people** struggle with the internet and **90%** have actually quit learning because of it, we know that standard apps just don't work here.

The main takeaway is that the app cannot just be a website on a phone. It has to be built for the worst-case scenario as the normal state. Students aren't just asking for videos; they are demanding the ability to **download work for later, save their progress automatically** so they don't lose data when the signal drops, and **use as little data as possible** because it's expensive.

Essentially, the app's success depends on it being light and reliable. If it doesn't buffer and it respects the student's storage and data balance, it wins. If it acts like a high-speed app built for Europe or the US, it will fail our users.